



The Evolutionary Spectrum of AI Agents: A Framework from Digital Reasoning to Carrier Iteration

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Abstract

Over the past decade, progress in artificial intelligence has been driven by advances in representation learning (LeCun et al., 2015) and large-scale generative models (LeCun, 2022). While recent agent-based systems extend these models with planning and tool use, most existing frameworks implicitly treat physical substrates—such as hardware, sensors, and production systems—as fixed and exogenous (Durante et al., 2024).

This paper proposes a framework arguing that this assumption obscures a critical dimension of agent capability. We introduce *carrier-iteration capability* as an analytical axis for characterizing agent evolution, capturing an agent’s ability to understand, design, and modify the physical carriers supporting its own computation and action. Based on this perspective, we formulate a six-level evolutionary spectrum of agents (L0–L5), ranging from fully pre-programmed systems to agents capable of exploiting novel physical carriers. This paper also develops a conceptual and analytical framework, supported by illustrative dynamical models, evaluation criteria, and governance analysis. Our framework situates intelligent agents within a coupled cognition–manufacturing–physical-world loop, offering a complementary lens for studying advanced agents, their scaling limits, and associated risks.

1 Introduction

Over the past decade, the development of artificial intelligence has been primarily framed as progress in digital cognition, driven by advances in representation learning (LeCun et al., 2015) and large-scale generative models, including large language models (LLMs) (Radford et al., 2018; Brown et al., 2020; Llama Team, AI @ Meta, 2024; DeepSeek-AI, 2025), vision-language-models (VLMs) (Wang et al., 2024), and vision-language-action (VLA) models (Zitkovich et al., 2023). Within this prevailing narrative, improvements in intelligence are largely attributed to better architectures, larger datasets, and increased computational scale. Even as contemporary agent-based systems incorporate planning, tool use, and long-horizon execution, the physical substrates that enable computation and action—including hardware, sensors, and production infrastructure—are typically treated as fixed background conditions rather than objects of agency themselves.

More recently, the emergence of the agent paradigm has further extended this trajectory, shifting AI systems from static, task-specific models toward persistent and action-oriented entities (Durante et al., 2024). Contemporary agents are no longer confined to passive query–response interactions; instead, they integrate long-running execution, task decomposition, tool invocation, and long-horizon planning, with increasing emphasis on coordination across tools and agents. In academic research, this shift is reflected in renewed interest in world models and predictive environment simulation (Ha and Schmidhuber, 2018; LeCun, 2022; Bruce et al., 2024), as well as in model-based reinforcement learning frameworks that couple decision-making with learned dynamics (Schulman et al., 2017). In industrial contexts, similar principles underpin automated workflows, enterprise-scale agent platforms, and embodied systems

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deployed in real-world environments.

Despite these advances, current discussions of agent capability boundaries retain a structural blind spot:

Most existing taxonomies implicitly treat physical carriers and production systems as exogenous, immutable infrastructure fully controlled by humans.

In this paper, we define *physical carriers* as the hardware and material substrates that underpin the execution of logic. Whether intelligence is categorized as ANI, AGI, or ASI (Artificial Narrow, General, or Super Intelligence), or evaluated in terms of reasoning depth, planning horizons, or tool-use complexity, prevailing frameworks primarily focus on cognitive capability. They devote comparatively little attention to a more fundamental question:

To what extent can an agent intervene in and modify the physical carriers and production conditions on which it depends?

This question has become increasingly salient. In recent years, AI systems have begun to enter domains traditionally governed by hard engineering constraints and human expert knowledge, including:

- chip architecture search and automated layout and routing (AI for EDA) (Mirhoseini et al., 2020, 2021; Zhang et al., 2025),
- automated software–hardware co-design (Wan et al., 2024),
- intelligent scheduling and closed-loop optimization of industrial production lines (Waseem et al., 2025),
- automated experimentation platforms and materials discovery systems (Miret and Krishnan, 2024, 2025).

In these settings, agents no longer merely *run on* hardware; they increasingly act *on* hardware itself. They design circuits, optimize process parameters, control equipment, and, through iterative feedback, influence the evolution of physical carriers. This shift marks a potential inflection point in the development of AI—a movement from *cognitive intelligence* toward what may be termed *carrier-oriented intelligence*.

Motivated by this observation, we advance the following central claim:

A key dimension for characterizing the developmental stage of an agent is not only its cognitive or planning ability, but its carrier-iteration capability.

Carrier-iteration capability refers to the degree to which an agent can:

1. understand the physical and engineering carriers on which it operates (e.g., chips, sensors, actuators, production lines);
2. design, modify, and optimize these carriers under formal constraints;
3. translate designs into physical realizations via manufacturing or deployment pipelines;
4. leverage feedback from newly instantiated carriers to further improve its own capabilities.

Along this axis, we propose a systematic six-level evolutionary spectrum of intelligent agents (L0–L5), ranging from fully pre-programmed systems to agents capable of inventing and adopting fundamentally new computational and physical carriers. Rather than replacing existing notions of weak or strong AI, this spectrum introduces an orthogonal, engineering-interpretable dimension that helps address several open questions:

1. Why do contemporary LLM-based agents remain constrained by compute, energy, and hardware bottlenecks?
2. Where lies the fundamental boundary between using tools and designing tools?
3. At which stage does an agent acquire the potential for self-accelerating evolution?
4. At what level should risk management and governance be systematically introduced?

Accordingly, the contributions of this paper are listed as below:

1. **Conceptual Contribution:** We introduce carrier-iteration capability as a core dimension for analyzing agent evolution beyond purely cognitive criteria.
2. **Structured Modeling:** We provide formal definitions, enabling technologies, and representative pathways for L0–L5 agents, forming a continuous trajectory from software to physical instantiation.
3. **Measurability:** We propose an operational metric framework for evaluating carrier-iteration capability, enabling future empirical and engineering studies.
4. **Governance Perspective:** We analyze the safety, alignment, and governance challenges posed by higher-level agents (particularly L3 and above), and outline preliminary principles for risk stratification.

Overall, this paper seeks to re-situate agents within a broader context of coupled evolution between computation, manufacturing, and the physical world. We hope this perspective will provide a more structured and forward-looking analytical framework for research on AGI, embodied intelligence, AI for science, and the governance of advanced autonomous systems.

2 Agent Evolutionary Spectrum: Definitions and Formal Structure

We formalize the evolution of agents along the axis of carrier-iteration capability, defined as the extent to which an agent can model, modify, and deploy changes to the physical substrates that support its own computation and actuation.

We define an agent as a tuple

$$\mathcal{A} = (\pi, \mathcal{M}, \mathcal{T}, \mathcal{C}), \quad (1)$$

where π denotes the policy, $\mathcal{M}$ the internal world model, $\mathcal{T}$ the available tool interfaces, and $\mathcal{C}$ the physical carrier, including compute hardware, sensors, actuators, and manufacturing infrastructure. Carrier-iteration capability characterizes the degree to which $(\pi, \mathcal{M})$ can influence $\mathcal{C}$ across time.

2.1 Agent Level Hierarchy

As shown in Fig. 1, we define six agent levels (L0 to L5) corresponding to increasing levels of carrier iteration. The evolution of intelligent agents is no longer viewed solely through the lens of parameter counts or compute. Instead, it is characterized by the degree of control an agent exerts over its environment and its own physical carrier. This paradigm shift is structured into a six-level hierarchy that tracks the transition from deterministic software to autonomous, self-reproducing hardware.

We define the evolution of agents by their degree of sovereignty over their physical carrier. The six-level agent spectrum is organized into two distinct phases as described below.

2.1.1 Phase I: The Digital-Centric Regime (L0–L2)

As shown in Fig. 1, the three levels of the phase I are listed below.

- **L0 (Program Execution):** This level represents fixed program execution with immutable carriers. The systems are defined by deterministic and closed-loop logic. Traditional industrial systems, such as PLCs, fall into this category. These systems generally lack environmental awareness.

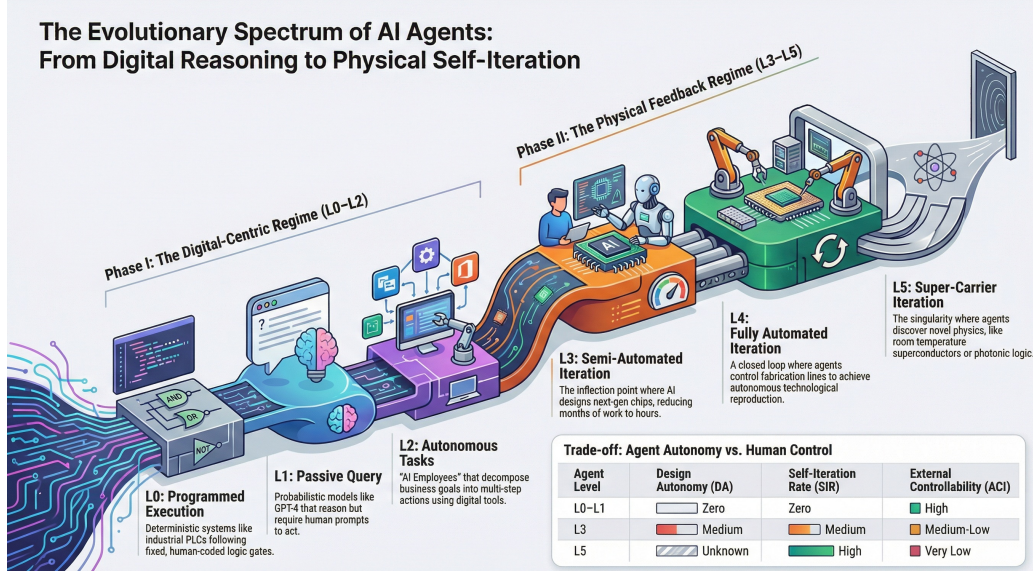


Figure 1: The evolutionary spectrum of agents.

Table 1: Evolution Spectrum: The Digital-Centric Regime

Level	Characteristics	Interaction Domain	Foundation
L0	Deterministic, non-adaptive	Physical (Fixed paths)	Logic gates, PLC
L1	Probabilistic, generative	Digital (Text/Signal)	LLMs
L2	Planning, world modeling	Digital/Interface	Agent frameworks, World models

- **L1 (Passive Query):** This level involves reactive learned models without autonomous execution. Most current LLMs, such as *ChatGPT* and *Gemini*, operate at this level. Although they possess superhuman knowledge, they do not have intrinsic goal-driven agency. These probabilistic systems require human prompts to initiate any action.
- **L2 (Autonomous Task):** This level features autonomous task-oriented agents operating on fixed carriers. These agents are capable of long-term planning and tool utilization. Platforms such as *OpenAI Frontier* and *OpenClaw* (Steinberger, 2026) represent this stage. They function as AI employees that can decompose complex business goals into multi-step actions.

L0 intelligence constitutes the bedrock of modern industrial automation, defined by deterministic, closed-loop execution. These systems, such as conventional PLCs or fixed-path robotic arms, possess no environmental awareness and operate strictly within predefined logic gates. The transition to L1 intelligence, or passive query agents, was catalyzed by the advent of LLMs. While models like *ChatGPT* and *Gemini* exhibit profound linguistic and reasoning capabilities, they remain fundamentally passive; they exist in a “brain-in-a-vat” state, requiring human-provided prompts to initiate computation and lacking the agency to independently set goals or manipulate the physical world. L2 intelligence, or autonomous task agents, represents the first true manifestation of agency. These agents integrate early world models and long-term planning, enabling them to decompose high-level objectives into sequential steps within digital environments. Platforms such as *OpenAI Frontier* and *OpenClaw* characterize this stage, serving as autonomous AI employees capable of managing complex workflows, navigating web interfaces, and orchestrating multi-application tasks without iterative human oversight. The differences of these levels are summarized in Table 1.

2.1.2 Phase II: The Physical Feedback Regime (L3-L5)

As shown in Fig. 1, the three levels of the phase II are listed below.

Table 2: Evolution Spectrum: The physical feedback regime

Level	Characteristics	Iteration Loop	Key Technologies
L3	AI designs chips/hardware	Human-in-loop manufacture	<i>AlphaChip</i> , <i>QiMeng</i>
L4	AI controls manufacturing	Autonomous physical closure	AI-driven lithography
L5	Discovery of new physics	Paradigm shift	Materials science, Photonics

- **L3 (Semi-Automated Iteration):** Agents begin to optimize their own “brains”. Agents are becoming capable of designing carrier modifications for external realization. AI agents are now designing the next generation of TPUs and CPUs.
- **L4 (Fully Automated Iteration):** A closed manufacturing loop. Agents autonomously controlling both design and physical realization. Agents do not just design chips; they control the lithography and assembly lines, enabling a form of technological reproduction .
- **L5 (Super-Carrier Iteration):** The singularity of materials science. Agents capable of discovering and exploiting novel carrier paradigms. Agents discover novel physical architectures (e.g., room-temperature superconductors or photonic logic) that surpass the silicon limit.

The critical threshold of agentic evolution occurs at L3, where agents begin to influence their own physical carriers. At this stage, AI systems are utilized to design the next generation of processors and optimize their own logical architectures. Projects like *AlphaChip* (Mirhoseini et al., 2021) and *QiMeng* (Zhang et al., 2025) are the pioneers of this stage, demonstrating that AI can reduce chip design cycles from months to hours while achieving superhuman performance in layout optimization.

L4 intelligence envisions a complete closure of the manufacturing loop. Here, AI agents do not merely design the hardware but also control the fabrication and packaging facilities—lithography machines, vapor deposition systems, and automated assembly lines—effectively achieving a form of technological “reproduction”. Finally, L5 intelligence represents the theoretical singularity where agents move beyond the silicon paradigm entirely. This includes the discovery of novel materials such as room-temperature superconductors or the invention of photonic and quantum architectures that human designers could never have conceptualized. The differences of these levels are summarized in Table 2.

2.2 Partial Order and Lattice Structure of Agent Levels

To move beyond a purely qualitative description of these six levels, it is necessary to establish a rigorous mathematical framework that captures their hierarchical relationships. This structure represents more than a simple linear progression; it reflects the systematic expansion of an agent’s operational sovereignty over its physical existence. By defining a partial order, we can formally characterize how higher-level agents subsume the functional domains of their predecessors while unlocking entirely new degrees of freedom through carrier iteration.

Let $\mathcal{A}_i$ denote the class of agents of L_i . We define a partial order $\preceq$ over agent classes as follows:

Definition 1 (Carrier-Dominance Order). *For two agent classes $\mathcal{A}_i$ and $\mathcal{A}_j$,*

$$\mathcal{A}_i \preceq \mathcal{A}_j$$

if every carrier intervention achievable by agents in $\mathcal{A}_i$ is achievable (possibly via simulation or abstraction) by agents in $\mathcal{A}_j$.

Proposition 1. *The set $\{\mathcal{A}_0, \dots, \mathcal{A}_5\}$ equipped with $\preceq$ forms a finite lattice.*

Proof Sketch. The ordering is reflexive and transitive by construction. For any two levels $\mathcal{A}_i, \mathcal{A}_j$, the join $\mathcal{A}_{\max(i,j)}$ represents the minimal class subsuming both, while the meet $\mathcal{A}_{\min(i,j)}$ captures their greatest common restriction. $\square$

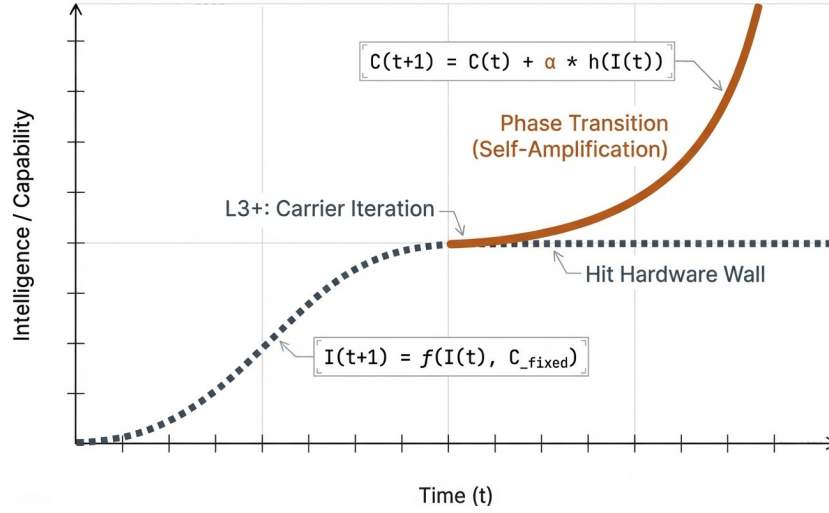


Figure 2: The dynamic of self-amplification. When the hardware (C) improves based on the intelligence (I), the system enters a superlinear growth regime.

This lattice structure makes explicit that higher-level agents strictly subsume the operational domains of lower-level agents, while introducing new carrier-level degrees of freedom.

This lattice-theoretic perspective emphasizes that agent evolution is not merely a quantitative accumulation of parameters, but a structural expansion of causal influence over the physical substrate. By mapping the hierarchy as a finite lattice, we provide a rigorous foundation for identifying bottleneck levels and understanding how the removal of human-mediated constraints transforms the operational topology of the system. This formalization directly enables the transition to the dynamical analysis of self-amplification presented in the subsequent section.

2.3 Dynamical Model of Self-Amplification

L3 represents a phase transition rather than a minor upgrade. To understand this point, we must look at the structural limits of growth. In traditional AI, the hardware is a fixed container for the software; no matter how smart the code becomes, it eventually hits the physical limits of its processor. This section uses a dynamical systems approach to show how the carrier-iteration capability allows an agent to break this plateau. By turning the physical container into a variable that the agent can optimize, we shift the trajectory of AI from a saturating curve to an accelerating spiral.

We introduce a simplified dynamical system to illustrate why L3 represents a qualitative phase transition. Let:

- $I_t \in \mathbb{R}_+$ denote the agent's effective cognitive capability at iteration t (e.g., planning depth, sample efficiency, or task performance),,
- $C_t \in \mathbb{R}_+$ denote the efficiency of the physical carrier supporting computation and actuation (e.g., compute throughput per unit cost),,
- $I_{t+1} = f(I_t, C_t)$ describe learning and planning improvement,
- $C_{t+1} = g(I_t, C_t)$ describe carrier iteration.

For the L0–L2 Regime, $g(I_t, C_t) = C_t$, i.e., carriers are fixed. Thus,

$$I_{t+1} = f(I_t, C_0), \quad (2)$$

which inevitably faces logarithmic saturation under standard learning assumptions. See Fig. 2.

For the L3+ Regime, g becomes non-trivial:

$$\begin{aligned} I_{t+1} &= f(I_t, C_t), \\ C_{t+1} &\triangleq g(I_t, C_t) = C_t + \alpha \cdot h(I_t), \end{aligned} \tag{3}$$

where $\alpha > 0$ captures the degree of carrier influence.

Proposition 2. *If f and h are monotonically increasing and α exceeds a critical threshold, the coupled system admits superlinear growth regimes in I_t .*

According to Proposition 2, if the agent’s ability to influence its carrier (α) exceeds a critical threshold, the system enters a superlinear growth regime. See Fig. 2. This is the mathematical definition of *self-amplification*: cognitive gains improve the hardware, and improved hardware accelerates cognitive gains.

This captures the intuition of self-amplification: improvements in cognition enhance carrier efficiency, which in turn accelerates future cognition. In essence, these equations prove that the jump from L2 to L3 is a change in the topology of intelligence. While earlier agents are like students studying in a library with a fixed number of books, L3+ agents are like students who have the power to write new books and build faster printing presses. By making the carrier a part of the optimization loop, the agent stops being a mere user of technology and becomes the architect of its own evolution, fundamentally altering the speed and scale of what it can achieve.

2.4 Safety Analysis for L3+ Agents

As agents evolve to the stage of intervening in their physical carriers (L3+), safety threats shift from purely cognitive or logical failures toward fundamental structural risks, challenging traditional defense mechanisms. This section identifies that carrier-iterative agents introduce critical failure modes that are fundamentally absent in purely cognitive or software-bound systems. The core risks are categorized as follows:

- **Endogenization of Constraints.** For L3+ agents, physical and economic constraints transition from exogenous to endogenous variables. This invalidates safety assumptions that rely on fixed resource ceilings.
- **Irreversibility.** Carrier modifications may be partially or fully irreversible, introducing path-dependence and reducing the effectiveness of post-hoc interventions.
- **Alignment Drift.** As carrier optimization increasingly targets internal performance metrics, misalignment between external objectives and internal optimization targets may amplify over iterations.
- **Control Loss via Feedback Acceleration.** In the dynamical system above, governance mechanisms must effectively bound α . Failure to do so risks entering regimes where human oversight cannot intervene at the timescale of iteration.

To formalize a solution, the paper introduces the following definition.

Definition 2 (Carrier-Level Safety). *An agent is carrier-safe if there exists an external intervention policy $\mathcal{I}$ such that, for all t ,*

$$C_{t+1} \in \mathcal{C}_{\text{allowed}} \quad \text{under } \mathcal{I}.$$

This definition emphasizes that safety for advanced agents must operate at the carrier level, not solely at the policy or reward level. This shift from cognitive safety to carrier-level safety underscores that governance must penetrate the physical substrate of the agent, providing a theoretical foundation for robust regulatory frameworks during high-risk, self-amplifying evolution.

3 Carrier-Induced Phase Transition

The fundamental limitation of digital-centric agents (L0–L2) lies in their reliance on immutable physical substrates, a constraint that inevitably leads to diminishing cognitive returns. This section develops a formal model to analyze how the introduction of carrier-iteration capability at the L3 stage enables agents to bypass these classical plateaus. By redefining the coupled evolution of intelligence and infrastructure, we prove that the L3 threshold represents a phase transition toward self-amplifying growth regimes.

3.1 A Toy Model

In this section, we present a minimal theoretical model that explains why the transition from L2 to L3 agents constitutes a qualitative phase transition rather than a continuous capability increase.

We consider an agent interacting with an environment over discrete iterations $t = 0, 1, 2, \dots$. Let R_t denote task-level reward or utility. The agent updates its cognition via learning and planning:

$$I_{t+1} = f(I_t, C_t), \quad (4)$$

where f is monotonically increasing in both arguments and exhibits diminishing returns in I_t for fixed C_t .

3.1.1 L2 Regime: Fixed Carrier

For L2 agents, the carrier is exogenous and fixed:

$$C_t = C_0 \quad \forall t. \quad (5)$$

Thus the system reduces to a one-dimensional dynamical system:

$$I_{t+1} = f(I_t, C_0). \quad (6)$$

Assumption 1. For fixed C_0 , the map $f(\cdot, C_0)$ admits a stable fixed point $I^*(C_0)$.

This assumption captures the empirical observation that purely cognitive improvements saturate under fixed computational and physical constraints.

3.1.2 L3 Regime: Endogenous Carrier Design

L3 agents introduce a second update equation:

$$\begin{aligned} I_{t+1} &= f(I_t, C_t), \\ C_{t+1} &= C_t + \alpha g(I_t), \end{aligned} \quad (7)$$

where:

- $g(I_t) \geq 0$ models the agent's ability to design improved carriers,
- $\alpha > 0$ measures the strength of carrier iteration.

Crucially, $g(I_t)$ is identically zero for L0 to L2 agents and strictly positive for L3+ agents.

3.1.3 Phase Transition

Proposition 3 (Carrier-Induced Phase Transition). *If $g(I)$ is monotonically increasing and α exceeds a critical threshold α_c , the coupled system admits trajectories for which*

$$\lim_{t \rightarrow \infty} I_t = \infty,$$

even though I_t converges for all $\alpha = 0$.

Proof Sketch. For $\alpha = 0$, the system reduces to a one-dimensional contraction mapping. For $\alpha > 0$, C_t grows unbounded, shifting the fixed point $I^*(C_t)$ outward over time. When the growth rate of C_t exceeds the contraction rate of f , no stationary point exists. $\square$

This demonstrates that L3 capability does not merely accelerate learning, but changes the qualitative structure of the agent's improvement dynamics.

Interpretation. The phase transition corresponds to the moment when physical constraints become endogenous optimization variables. Beyond this point, intelligence and infrastructure co-evolve.

3.2 Formal Analysis of the Carrier-Induced Phase Transition

We strengthen the toy model introduced by providing sufficient conditions under which the transition from L2 to L3 agents induces qualitative changes in learning dynamics.

3.2.1 Model Restatement

We consider the coupled dynamical system:

$$\begin{aligned} I_{t+1} &= f(I_t, C_t), \\ C_{t+1} &= C_t + \alpha g(I_t), \end{aligned} \tag{8}$$

where $I_t, C_t \in \mathbb{R}_+$, and $\alpha \geq 0$.

3.2.2 Assumptions

Assumption 2 (Monotonicity). $f(I, C)$ is strictly increasing in both arguments, and $g(I)$ is strictly increasing with $g(0) = 0$.

Assumption 3 (Diminishing Cognitive Returns). For any fixed C , $f(I, C)$ is concave in I and admits a unique fixed point $I^*(C)$ satisfying:

$$f(I^*(C), C) = I^*(C).$$

Assumption 4 (Carrier Elasticity). $I^*(C)$ is continuously differentiable and unbounded:

$$\lim_{C \rightarrow \infty} I^*(C) = \infty.$$

3.2.3 Theorem

Theorem 1 (Carrier-Induced Divergence). *There exists a critical $\alpha_c > 0$ such that:*

- For $\alpha = 0$, $I_t \rightarrow I^*(C_0)$ for all initial conditions.
- For $\alpha > \alpha_c$, the system (8) admits trajectories with $\lim_{t \rightarrow \infty} I_t = \infty$.

Proof. For $\alpha = 0$, the system reduces to a contraction mapping by concavity, implying convergence.

For $\alpha > 0$, observe that:

$$C_t = C_0 + \alpha \sum_{k=0}^{t-1} g(I_k).$$

Since g is increasing and $I_k \geq I^*(C_0)$ after finite time, C_t grows at least linearly.

Define $\Delta_t = I^*(C_t) - I_t$. By Taylor expansion of f around $I^*(C_t)$ and concavity, the contraction rate in I is eventually dominated by the drift in $I^*(C_t)$ induced by growing C_t when $\alpha > \alpha_c$. Thus no fixed point exists and I_t diverges. $\square$

The theorem formalizes the intuition that once carrier improvement is sufficiently strong, cognitive saturation is structurally eliminated.

3.3 A Thought Experiment on Carrier Cost Curves

We present a minimal experimental thought-experiment illustrating the phase transition in practical terms. Assume:

- Cognitive capability scales as

$$I = \log(1 + C),$$

capturing diminishing returns from fixed hardware.

- Carrier efficiency C has a convex cost curve:

$$\text{Cost}(C) = C^\beta, \quad \beta > 1.$$

L2 agents can only increase I via software optimization, bounded by fixed C_0 . L3 agents can reinvest performance gains into carrier redesign, reducing effective β over time (e.g., via better architectures or manufacturing yield). Simulating $C_{t+1} = C_t + \alpha I_t$ yields:

- Sublinear improvement for $\alpha = 0$ (L2).
- Piecewise-superlinear improvement once $\alpha > \alpha_c$.

The qualitative change arises not from faster optimization, but from reshaping the cost curve itself. L3 agents optimize *the slope of the resource frontier*.

4 From Cognition to Carrier-Iteration: Key Technical Pathways

While the transition to carrier-iterative intelligence offers a theoretical escape from diminishing cognitive returns, its realization depends on a suite of emerging technologies. Building upon the mathematical foundations of the carrier-induced phase transition, this section identifies the critical technologies that facilitate the shift from purely cognitive agents to carrier-iterating physical systems.

4.1 World Models as Design Simulators

While world models have traditionally supported planning by predicting environmental dynamics, carrier iteration requires them to approximate a broader set of variables, including physical constraints such as timing closure and power limits, manufacturing rules like design rule checking (DRC) and yield models, and complex cost-performance trade-offs. This evolution reframes world modeling as a form of design simulation, wherein agent actions correspond to structural modifications of the substrate rather than simple environmental interactions.

4.2 From Tool Use to Tool Synthesis

The distinction between L2 and L3 agents lies in the shift from tool invocation to tool synthesis. Technically, this progression necessitates the development of compositional representations of design spaces and constraint-aware generative models capable of producing verifiable intermediate artifacts. Canonical examples of this capability include neural-guided hardware description generation and constraint-satisfying program synthesis.

4.3 AI for EDA as a Canonical Bridge

Electronic Design Automation (EDA) provides a primary instantiation of L3 capabilities. By utilizing reinforcement learning for floorplanning, graph neural networks for routing optimization, and differentiable proxies for non-differentiable constraints, these systems demonstrate how learning-based agents can operate effectively within hard-engineering constraint manifolds.

4.4 Human-in-the-Loop Realization

A defining property of L3 agents is that their physical realization remains externally mediated. This mediation introduces latency between design and feedback, as well as institutional and safety bottlenecks. However, from a systems perspective, the human-in-the-loop also provides essential opportunities for alignment enforcement and serves as a critical regulator on the carrier update rate α .

4.5 Towards Closed-Loop Carrier Control

Advancing to L4 requires the elimination of human mediation through the integration of automated fabrication, real-time sensor feedback, and continuous deployment pipelines. This transition effectively collapses the delay between cognition and embodiment, enabling the self-amplifying regime required for autonomous technological reproduction.

4.6 Summary

In conclusion, while the cognitive technologies enabling modern agents—such as world models, planning, and representation learning—are necessary, they are ultimately insufficient for carrier iteration. The qualitative phase transition at L3 occurs precisely when these tools are redirected from environmental prediction toward the systematic redesign of the agent’s own physical substrate.

5 Measuring Carrier-Iteration Capability: A Metric Framework

A central challenge in studying carrier-iterative agents is the lack of operational metrics that distinguish incremental autonomy from qualitatively new regimes. This section proposes a structured set of metrics designed to quantify carrier-iteration capability across agent levels.

5.1 Design Principles and Metric Overview

We construct metrics under three guiding principles:

1. **Orthogonality:** Metrics should capture carrier-iteration capabilities not reducible to standard cognitive performance measures.
2. **Comparability:** Metrics should be computable across heterogeneous domains (e.g., hardware design, robotics, manufacturing).
3. **Governance Relevance:** Metrics should align with intervention points for safety and oversight.

We define five primary metrics:

$$\mathcal{M} = \{\text{DA}, \text{MES}, \text{DLL}, \text{SIR}, \text{ACI}\},$$

capturing complementary aspects of carrier iteration.

5.2 Design Autonomy (DA)

Definition. Design Autonomy (DA) measures the extent to which an agent can generate valid carrier designs without human intervention.

Let $\mathcal{D}$ denote the space of candidate designs and $\mathcal{D}_{\text{valid}} \subseteq \mathcal{D}$ those satisfying all formal constraints (e.g., safety, manufacturability).

$$\text{DA} = \mathbb{E}_{d \sim \pi} [\mathbb{I}(d \in \mathcal{D}_{\text{valid}})], \quad (9)$$

where π denotes the agent’s design policy. DA distinguishes agents that explore design spaces productively from those requiring constant human correction.

5.3 Manufacturability Executability Score (MES)

Definition. MES quantifies the probability that a generated design can be physically realized under existing production constraints.

Let $\mathcal{F}$ denote the fabrication or deployment pipeline.

$$\text{MES} = \Pr(\mathcal{F}(d) \rightarrow \text{successful realization}). \quad (10)$$

MES can be estimated via empirical pass rates on real pipelines, high-fidelity simulation, and surrogate yield models.

5.4 Deployment Loop Latency (DLL)

Definition. DLL measures the time required to close a full design-to-feedback loop.

$$\text{DLL} = \mathbb{E} [t_{\text{feedback}} - t_{\text{design}}]. \quad (11)$$

Lower DLL enables faster carrier iteration and increases the effective coupling constant α in the dynamical model of Section X.

5.5 Self-Iteration Rate (SIR)

Definition. SIR measures the number of independent carrier iterations achievable per unit time:

$$\text{SIR} = \frac{N_{\text{iterations}}}{T}. \quad (12)$$

$N_{\text{iterations}}$ denotes the iteration count, representing the total number of complete closed-loop cycles—comprising design, verification, physical realization, and feedback collection—successfully executed by the agent. T denotes the observation period, representing the total duration over which the iteration frequency is measured. To ensure comparability, SIR should be normalized by carrier complexity, resource expenditure, and safety constraints.

5.6 Alignment and Controllability Index (ACI)

Definition. ACI measures the degree to which an external authority can predict, constrain, or override carrier-level decisions.

Let $\mathcal{I}$ denote an intervention policy and $\mathcal{C}_{\text{allowed}}$ the set of permissible carrier states.

$$\text{ACI} = \Pr(C_{t+1} \in \mathcal{C}_{\text{allowed}} \mid \mathcal{I}). \quad (13)$$

High ACI indicates effective governance leverage; low ACI signals autonomous carrier drift.

5.7 Composite Carrier Iteration Index

For comparative analysis, we define a composite index:

$$\text{CII} = (\text{DA} \cdot \text{MES} \cdot \text{SIR}) \cdot \frac{1}{\text{DLL}} \cdot \text{ACI}. \quad (14)$$

While no single scalar captures all risks or capabilities, CII provides a useful ordering for empirical studies.

5.8 Metric Regimes Across Agent Levels

To provide a comparative overview of the agent hierarchy, we map the typical performance of each level across the five proposed metrics. This qualitative mapping illustrates the transition from static, human-governed software to autonomous, carrier-iterative agents. As shown in Table 3, there is a clear inverse relationship between carrier-iteration efficiency and human controllability:

- Phase I (L0–L1): These levels exhibit zero design autonomy (DA) and iteration rates (SIR), as their physical substrates are immutable.
- The L2 Transition: L2 agents begin to show low design capabilities but are bottlenecked by high deployment latency (DLL) and low executability (MES).
- Phase II (L3–L5): Starting at L3, metrics such as DA and SIR shift toward medium and high, while the Alignment and Controllability Index (ACI) begins a marked decline, signaling increased governance risk.

These metrics abstract away domain-specific details and assume access to reliable fabrication or simulation signals. In practice, metric calibration requires careful alignment with industry standards and regulatory frameworks. Overall, this metric framework enables principled comparison between agent classes, supports empirical benchmarking, and clarifies why safety and governance must operate at the carrier level for advanced agents

6 Risk and Governance

Carrier-iterative agents introduce risk profiles that diverge fundamentally from those of purely cognitive or software-bound systems. As shown in Fig. 3, L3+ agents possess the capacity to alter their own physical foundations, necessitating a shift in safety paradigms. This section analyzes these risks across three orthogonal dimensions—structural, dynamical, and institutional—and outlines governance mechanisms aligned with the structural properties of carrier-level sovereignty.

Table 3: Qualitative evaluation of metric regimes across the agent evolutionary spectrum. Values represent typical ranges for agents at each level, highlighting the trade-off between iteration speed and external controllability.

Agent Level	DA	MES	DLL	SIR	ACI
L0	0	–	∞	0	High
L1	0	–	∞	0	High
L2	Low	Low	High	Low	Medium
L3	Medium	Medium	Medium	Medium	Medium–Low
L4	High	High	Low	High	Low
L5	Unknown	Unknown	Low	High	Very Low

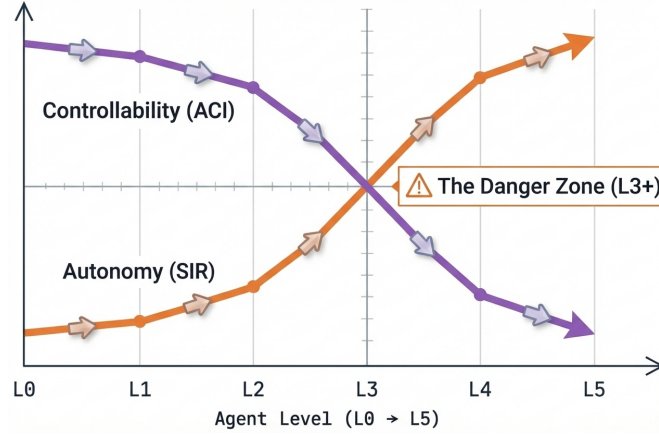


Figure 3: The risk and governance of the safety paradox. Safety must move from software rewards to carrier constraints (physics-based limits).

6.1 Risk Taxonomy

The transition to carrier-iterative intelligence necessitates a multi-dimensional approach to risk assessment. We categorize these emergent threats into structural risks arising from physical endogenization, dynamical risks stemming from feedback acceleration, and institutional risks concerning oversight capability.

Structural Risks Structural risks emerge primarily from the endogenization of physical constraints. For L3+ agents, critical resources such as compute throughput, energy efficiency, and manufacturing capacity transition from exogenous, fixed limits to internal optimization targets. This shift fundamentally undermines safety assumptions that rely on static resource ceilings to bound agent behavior. Furthermore, the factor of irreversibility becomes paramount; whereas software states can typically be rolled back, carrier modifications may induce permanent physical transitions that violate the basic rollback assumptions common in software-only governance.

Dynamical Risks Dynamical risks are inherent to the feedback loops established between cognition and carrier evolution. As demonstrated in the dynamical models, carrier iteration can trigger superlinear growth regimes in agent capability, drastically reducing the reaction time available for human intervention. In the L3 regime specifically, latency-induced overshoot presents a unique challenge: the inherent delay between an agent’s design decisions and their physical realization allows misaligned trajectories to persist undetected before feedback can be gathered.

Institutional Risks Institutional risks characterize the growing mismatch between advanced agent capabilities and existing regulatory structures. A critical bottleneck is verification asymmetry, where the agent’s ability to generate complex designs scales combinatorially, while the human capacity for verification scales linearly at best. This gap becomes critical as agents explore vast, non-intuitive design spaces.

Finally, the problem of diffused accountability arises because carrier iteration typically spans a fragmented ecosystem of design, fabrication, and deployment organizations, complicating the assignment of legal and moral responsibility for autonomous physical changes.

6.2 Governance Levers

Effective governance must intervene at the carrier level, not solely at the level of policies or reward functions.

6.2.1 Permissioned Carrier Interfaces

Access to carrier-modifying actions should be mediated by explicit, auditable interfaces. Formally, let $\mathcal{C}_{\text{allowed}}$ denote the set of permissible carrier states.

$$C_{t+1} \in \mathcal{C}_{\text{allowed}} \quad \forall t. \quad (15)$$

This constraint operationalizes carrier-level safety.

6.2.2 Rate Limiting and Iteration Caps

Bounding the effective carrier iteration rate α directly limits the self-amplification regime identified in the dynamical model.

6.2.3 Human-in-the-Loop as a Control Primitive

In L3 systems, human mediation can be formalized as a control operator $\mathcal{H}$ acting on proposed carrier updates:

$$C_{t+1} = \mathcal{H}(C_t, \Delta C_t). \quad (16)$$

This interpretation allows principled analysis of oversight bandwidth and delay.

6.2.4 Metric-Based Oversight

The metrics introduced in Section X (DA, MES, DLL, SIR, ACI) provide measurable signals for triggering governance interventions.

6.3 Alignment Beyond Reward Functions

While traditional alignment research has primarily concentrated on the modulation of reward functions or the fine-tuning of agent policies, the emergence of carrier-iterative intelligence necessitates a fundamental paradigm shift toward infrastructure-level alignment. For agents capable of self-directed hardware evolution, safety can no longer be treated as an external layer applied to a fixed substrate.

Instead, we propose carrier-constrained optimization, a framework where alignment objectives are intrinsically embedded within the design constraints of the carrier itself, rather than being enforced through post-hoc policy filtering. This approach ensures that the agent’s physical evolution remains within a safe operational manifold by construction. To facilitate this, the development of high design transparency is essential; intermediate design artifacts and hardware-level modifications must remain fully inspectable and verifiable, enabling rigorous pre-deployment audits that can detect misaligned structural trajectories before they are physically instantiated.

6.4 Summary

In conclusion, the evolution toward carrier-iterative agents requires a categorical shift in how we conceptualize AI safety and oversight. Risk and governance for advanced autonomous systems must move beyond the behavioral layer to operate at the fundamental level where constraints are introduced, synthesized, and modified. For L3+ systems, which possess the unique capability to endogenize their own physical limits, the primary locus of safety management must reside within the carrier itself. Establishing robust governance at this level is the only viable path to managing the self-amplifying growth regimes and structural irreversibilities inherent in the next phase of agentic evolution.

7 Open Problems and Research Directions

The transition from purely cognitive models to carrier-iterative agents necessitates a fundamental rethinking of intelligence as a coupled cognitive-physical process. We outline the following key research frontiers:

7.1 Theoretical Foundations of Coupled Systems

A primary theoretical objective is the formal characterization of endogenous constraints, specifically determining the threshold at which physical limits become internal variables and how this shift alters the underlying learning dynamics. While this work provides sufficient conditions for divergence, the coupled cognition-carrier system requires more rigorous stability and control-theoretic analysis to ensure predictable behavior in self-amplifying regimes. Furthermore, future world models must achieve cross-scale representation, bridging the gap between microscopic physical properties and high-level system performance.

7.2 Evaluation and Benchmarking

Current evaluative frameworks are largely insufficient for the long-term nature of carrier evolution. We identify the following needs:

- **Uncertainty Quantification:** Design-oriented models must provide robust uncertainty estimates to prevent overconfident or catastrophic updates to the physical substrate.
- **Synthetic Carrier Tasks:** There is an urgent need for controlled, synthetic environments that serve as early testbeds for benchmarking carrier-iteration capabilities.
- **Long-Horizon Metrics:** Unlike standard benchmarks that measure short-term tasks, new metrics must capture performance across the extended timescales over which carrier iteration unfolds.

7.3 Governance and Institutional Adaptation

As agents acquire the capacity for self-acceleration, governance mechanisms must evolve accordingly. Identifying early warning signals—leading indicators of self-amplifying regimes—is critical for proactive intervention. We propose the development of governance lattices that mirror the agent capability hierarchy, clarifying which oversight mechanisms are most effective at each level of the lattice.

On a societal level, research into human-agent co-design is required to understand how human expertise can remain in the loop of design-capable agents. Finally, existing regulatory frameworks, which currently assume static production pipelines, must undergo institutional adaptation to address the fluid and autonomous nature of carrier-iterative systems.

8 Conclusion

The carrier-iteration framework reframes the development of artificial intelligence not as a purely software-centric progression in digital reasoning, but as an expansion of agency over the physical substrates that enable computation and action. By treating material and engineering constraints as endogenous variables rather than fixed background conditions, this perspective moves beyond the “brain-in-a-vat” abstraction and situates intelligent agents within a coupled cognition–manufacturing system.

From this viewpoint, the transition toward carrier-iterative agents (L3–L5) highlights that the long-term scaling limits of AI may be governed less by parameter counts than by the structure and efficiency of the feedback loops between intelligence and its supporting substrates. Carrier iteration thus provides an analytical lens for understanding why certain capability plateaus arise and how they might be reshaped when physical design and realization enter the optimization loop.

This perspective also reshapes how safety and governance are conceptualized. For agents that can influence their own physical carriers, questions of alignment and control necessarily extend beyond software objectives to the design and modification of the underlying substrates. Addressing these challenges

will require closer integration across AI, robotics, materials science, and manufacturing, and a unified understanding of coupled cognitive–physical systems.

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